

$$\begin{pmatrix} 1 & 2 & -1 \\ -2 & 0 & 4 \\ 3 & 1 & 5 \end{pmatrix}, \quad \begin{pmatrix} -\frac{2}{21} & -\frac{11}{42} & \frac{4}{21} \\ \frac{11}{21} & \frac{4}{21} & -\frac{1}{21} \\ -\frac{1}{21} & \frac{5}{42} & \frac{2}{21} \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & -1 \\ -2 & 0 & 4 \\ 3 & 1 & 5 \end{pmatrix}^3$$

$$\det \begin{pmatrix} 1 & 2 & -1 \\ -2 & 0 & 4 \\ 3 & 1 & 5 \end{pmatrix} = 42$$

$$\begin{pmatrix} 3 & 2 & -1 \\ 4 & 0 & -3 \\ -2 & 1 & 7 \end{pmatrix}^T = \begin{pmatrix} 3 & 4 & -2 \\ 2 & 0 & 1 \\ -1 & -3 & 7 \end{pmatrix}$$

Starting the reduction.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ -9 & -4 & -8 & -2 & 1 \\ -6 & -3 & -15 & 3 & 0 \\ 2 & -3 & 7 & 3 & 2 \\ 4 & -6 & 14 & 6 & 4 \end{pmatrix}$$

Now using the pivot with value 3 at row 1 and column 1.

Subtract from row 2 the pivot row multiplied by -3.

Subtract from row 3 the pivot row multiplied by -2.

Subtract from row 4 the pivot row multiplied by 2/3.

Subtract from row 5 the pivot row multiplied by 4/3.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ 0 & -1 & -29 & 4 & 13 \\ 0 & -1 & -29 & 7 & 8 \\ 0 & -\frac{11}{3} & \frac{35}{3} & \frac{5}{3} & -\frac{2}{3} \\ 0 & -\frac{22}{3} & \frac{70}{3} & \frac{10}{3} & -\frac{4}{3} \end{pmatrix}$$

Now using the pivot with value -1 at row 2 and column 2.

Subtract from row 3 the pivot row multiplied by 1.

Subtract from row 4 the pivot row multiplied by 11/3.

Subtract from row 5 the pivot row multiplied by 22/3.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ 0 & -1 & -29 & 4 & 13 \\ 0 & 0 & 0 & 3 & -5 \\ 0 & 0 & 118 & -13 & -\frac{145}{3} \\ 0 & 0 & 236 & -26 & -\frac{290}{3} \end{pmatrix}$$

Now using the pivot with value 118 at row 4 and column 3.

Exchange of row 4 with row 3.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ 0 & -1 & -29 & 4 & 13 \\ 0 & 0 & 118 & -13 & -\frac{145}{3} \\ 0 & 0 & 0 & 3 & -5 \\ 0 & 0 & 236 & -26 & -\frac{290}{3} \end{pmatrix}$$

Subtract from row 5 the pivot row multiplied by 2.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ 0 & -1 & -29 & 4 & 13 \\ 0 & 0 & 118 & -13 & -\frac{145}{3} \\ 0 & 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Now using the pivot with value 3 at row 4 and column 4.

$$\begin{pmatrix} 3 & 1 & -7 & 2 & 4 \\ 0 & -1 & -29 & 4 & 13 \\ 0 & 0 & 118 & -13 & -\frac{145}{3} \\ 0 & 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

No pivot found in column 5.

Done. The rank of the matrix is 4.

$$A = \begin{pmatrix} \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{6} & \frac{1}{7} & \frac{1}{8} \\ \frac{1}{9} & \frac{1}{10} & \frac{1}{11} \end{pmatrix}$$

Pivot $1/3$ at $1, 1$.

Subtract from row 2 the pivot row multiplied by $1/2$.

Subtract from row 3 the pivot row multiplied by $1/3$.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & 0 & 1 \end{pmatrix} \quad U = \begin{pmatrix} \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ 0 & \frac{1}{56} & \frac{1}{40} \\ 0 & \frac{1}{60} & \frac{4}{165} \end{pmatrix}$$

Pivot $1/56$ at $2, 2$.

Subtract from row 3 the pivot row multiplied by $14/15$.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & \frac{14}{15} & 1 \end{pmatrix} \quad U = \begin{pmatrix} \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ 0 & \frac{1}{56} & \frac{1}{40} \\ 0 & 0 & \frac{1}{1100} \end{pmatrix}$$

Pivot $1/1100$ at $3, 3$.

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{3} & \frac{14}{15} & 1 \end{pmatrix} \quad U = \begin{pmatrix} \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ 0 & \frac{1}{56} & \frac{1}{40} \\ 0 & 0 & \frac{1}{1100} \end{pmatrix}$$

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 9 \\ 1 & 2 & 3 \\ 4 & 5 & 9 \end{array}\right), \quad \left(\begin{array}{ccc} 1 & 2 & 3 \\ a & b & c \\ 1 & 3 & 5 \\ 2 & 4 & 3 \end{array}\right)$$